

NAYSAN MUNJE

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EDUCATION

Bachelor of Mechanical Engineering

Carleton University - 3.7/4 GPA

2024 - 2028

Ottawa, ON

SHAD Canada

National STEM enrichment summer program for grade 10 and 11 students

Jul 2023

Toronto, ON

EXPERIENCE

NSERC Research Internship, Autonomous Vehicles

May 2025 – Aug 2025

Advanced Biomechatronics and Locomotion Lab, Carleton University

Ottawa, ON

- Engineered a vehicle-integrated **PCB data logger** for real-time high speed data capture, including GPS, IMU, and CAN bus signals for machine learning models with **ESP32** and **Raspberry Pi** infrastructure used in published **PhD Thesis**.
- Collected, processed and filtered video and sensor data with **MATLAB** and **Python** for use in **academic papers**, confirming **simulated data** to 90% accuracy.

Undergraduate Research Assistant, Autonomous Vehicles

Aug 2024 – Present

Advanced Biomechatronics and Locomotion Lab, Carleton University

Ottawa, ON

- Created AI face detection and automated driving algorithms in **MATLAB**.
- Integrated computer vision systems into ongoing autonomous driving research projects.

Mechanical Design Lead / Mentor

May 2022 – Feb 2025

FIRST Robotics Competition

Ottawa, ON

- Led a **20 member** student team in designing, prototyping, and manufacturing competition robots with **SolidWorks**.
- Developed large scale mechanisms, including a chain driven three stage elevator and robotic arms.
- Represented the team as **Robot Operator**, achieving a provincial finalist position and qualifying for the **World Championship** in Houston, TX.

PROJECTS

4 DoF Robotic Arm, Cycloidal Gearbox Joints

2026

Fusion 360, Nema 17 Motors, ESP32

- Designed and assembled 20:1 **cycloidal actuators** (~4.5 Nm) for a 40 cm arm with 1.5 kg lift.
- Includes 3D printed bearings, tight tolerance camshafts, compact cycloidal gearboxes, and eccentric cycloid base.
- Precise motion** through limit switch calibration, **controlled** through browser based 3D model over **wifi** with an ESP32.
- Driven by stepper motors repurposed from a broken 3D printer.

Hexapod Robot

2026

Fusion 360, ESP32

- 18 DoF walking made in under 3 weeks for \$160 USD using MG996R servo motors.
- Custom **3D printed legs** and chassis controlled over wifi via an **ESP32**.
- URDF 3D model** to simulate walking gaits with variable parameters.
- 3 **walking patterns**, 9 stationary motion patterns, manual control.

Die Roller PCB

2024

SolidWorks, KiCad, ATtiny

- Custom 2 layer **KiCad** board with Charlieplexed LEDs and soldered components. Battery powered with random rolling patterns.
- 3D printed enclosure designed in **SolidWorks** around the finished board.

Omnidirectional Rover

2026

ESP32, DC Motors

- Fabricated a three wheeled chassis with encoder motors, omni wheels, and Wi-Fi control.

TECHNICAL SKILLS

CAD / manufacture: SolidWorks, Fusion 360, 3D printing, CNC, enclosure design

Electronics prototyping: KiCad, soldering, Arduino (UNO, ESP32, ATtiny), Raspberry Pi

Software: Python, MATLAB, C